

ILE DE BREHAT, France

WMO: 071210

Lat: 48.856N

Lon: 3.004W

Elev: 36

StdP: 100.89

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	1.2	-4.9	2.5	1.1	-3.1	2.9	2.1	20.1	8.5	18.7	8.1	6.7	80	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	22.2	18.2	21.1	17.8	20.1	17.3	19.1	21.2	18.3	20.2	17.7	19.5	5.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	20.0	17.6	12.7	19.3	17.0	12.2	18.8	54.4	21.4	51.9	20.1	50.1	19.7	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.5	12.1	DB	-3.0	26.2	1.9	1.7	-4.4	27.4	-5.5	28.4	-6.5	29.4	-7.9	30.6	(4)
(5)			WB	-4.2	20.6	2.0	0.9	-5.7	21.2	-6.9	21.7	-8.0	22.2	-9.5	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	7.2	7.1	8.8	9.6	12.4	14.6	17.0	17.4	15.8	13.5	10.3	8.5
	DBStd	4.25	3.03	3.07	2.11	2.05	2.04	1.75	1.50	1.66	1.81	2.14	2.48	2.69
	HDD10.0	349	91	87	49	31	3	0	0	0	0	1	27	61
	HDD18.3	2376	344	315	297	262	183	113	48	38	78	149	242	306
	CDD10.0	1034	6	5	11	18	78	138	216	230	174	111	34	13
	CDD18.3	19	0	0	0	0	0	1	6	9	3	0	0	0
	CDH23.3	21	0	0	0	0	1	3	5	12	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	6.2	7.3	7.2	6.5	6.3	5.5	5.5	5.2	5.1	5.7	6.5	6.7	7.2
Precipitation	PrecAvg	828	87	78	54	64	56	48	49	50	44	84	102	101
	PrecMax	1037	164	155	159	174	120	111	127	98	130	176	200	224
	PrecMin	626	17	17	6	19	15	9	18	17	3	13	39	23
	PrecStd	97	43	37	33	36	24	27	26	23	28	37	43	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	14.5	15.7	18.0	20.1	22.2	23.6	24.9	22.4	20.5	16.1	14.7
		MCWB	12.1	12.8	13.0	14.0	16.3	17.7	18.5	19.4	18.1	16.8	14.4	13.1
	2%	DB	12.7	12.9	13.8	15.3	18.3	19.9	21.7	22.4	20.6	18.5	15.0	13.9
		MCWB	11.5	11.4	11.6	12.3	15.0	17.1	18.2	18.4	17.6	15.9	13.7	12.5
	5%	DB	12.1	12.0	12.7	13.7	16.9	18.2	20.5	21.2	19.5	17.2	14.2	13.1
		MCWB	11.1	10.6	11.0	11.6	14.4	15.7	17.7	17.9	16.9	15.1	13.1	12.0
	10%	DB	11.2	11.2	11.7	12.6	15.6	17.3	19.6	20.2	18.6	16.3	13.5	12.3
		MCWB	10.1	10.0	10.3	11.0	13.4	15.1	17.2	17.3	16.3	14.6	12.4	11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	13.2	13.3	14.3	17.0	18.5	20.0	20.2	19.1	17.4	14.9	13.9
		MCDB	13.2	14.0	15.2	17.1	19.2	21.0	22.1	23.1	20.9	19.3	15.6	14.5
	2%	WB	12.0	11.8	12.2	12.9	15.5	17.2	18.8	19.3	18.2	16.3	14.0	13.0
		MCDB	12.7	12.6	13.4	14.6	17.5	19.3	20.7	21.7	19.9	17.6	14.6	13.5
	5%	WB	11.2	10.9	11.2	11.9	14.6	16.2	18.1	18.5	17.6	15.6	13.4	12.2
		MCDB	12.0	11.7	12.3	13.3	16.4	17.8	19.9	20.5	19.1	16.8	14.1	13.1
	10%	WB	10.2	10.1	10.5	11.2	13.7	15.4	17.5	17.9	16.8	15.0	12.7	11.3
		MCDB	11.1	11.1	11.5	12.4	15.4	16.8	19.2	19.5	18.1	16.0	13.5	12.3
Mean Daily Temperature Range	5% DB	MDBR	3.3	3.6	4.1	4.2	4.6	4.4	4.6	4.8	4.1	3.8	3.4	3.3
		MCDBR	4.1	4.7	5.6	5.9	6.9	6.2	6.1	6.4	5.3	5.0	4.0	4.2
		MCWBR	3.7	4.1	4.2	3.9	4.2	3.6	3.4	3.3	3.1	3.3	3.5	3.7
	5% WB	MCDBR	3.9	4.4	5.1	5.5	6.4	6.0	5.5	5.6	4.9	4.5	3.7	3.9
		MCWBR	3.9	4.3	4.1	3.9	4.2	3.7	3.4	3.4	3.3	3.5	3.5	3.8
Clear-Sky Solar Irradiance	taub	0.333	0.349	0.378	0.390	0.395	0.389	0.387	0.377	0.366	0.359	0.336	0.326	
	taud	2.414	2.393	2.322	2.311	2.347	2.392	2.400	2.447	2.459	2.465	2.445	2.382	
	Ebn at Noon	720	796	829	858	867	873	868	862	836	776	711	669	
	Edh at Noon	67	85	106	118	119	115	112	103	92	78	64	61	
All-Sky Solar Radiation	RadAvg	0.93	1.70	2.83	4.43	5.29	5.80	5.56	4.69	3.67	2.11	1.15	0.78	
	RadStd	0.08	0.20	0.29	0.41	0.41	0.61	0.32	0.37	0.27	0.16	0.11	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page